

Shuvam Banerji Seal

Computational Chemist · AI/ML & Information Retrieval Researcher · DeepTech & AI-Fintech Co-Founder

BS-MS Dual Degree, Fifth Year (Chemistry Major, Computer Science Minor) – IISER Kolkata

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Research interests: density functional theory and electronic structure (Quantum ESPRESSO, Gaussian) · molecular dynamics (LAMMPS) · machine learning for the physical and life sciences (transformers, mixture-of-experts) · structural bioinformatics and RNA structure prediction · information retrieval and retrieval-augmented generation · formal verification in Lean 4 · AI safety and evaluation · information theory.

Master's Thesis (Ongoing)

Gray-Code and Karnaugh-Map Structure of Nucleic-Acid Sequences, Formalised in Lean 4 2026–

BS-MS Thesis, IISER Kolkata

Encoding DNA, RNA and protein sequences into binary sequence space and arranging the codewords on Gray-code orderings and Karnaugh maps, so that map adjacency is a single-symbol mutation. Every theorem about the resulting binary relations, symmetries and distance structure is machine-checked in **Lean 4** (Mathlib).

A Transformer Mixture-of-Experts Model for RNA Structure Prediction 2026–

BS-MS Thesis, IISER Kolkata

A sparse **Mixture-of-Experts transformer** for RNA secondary and tertiary structure prediction, routing structural motifs (helices, loops, pseudoknots) to specialised experts at fixed inference cost. Geometry-aware attention with distance-map and torsion-angle heads.

Publications

AgriIR: A Scalable Framework for Domain-Specific Knowledge Retrieval ECIR 2026

S. Banerji Seal, A. Poddar, A. Mishra, D. Roy. Six-stage configurable RAG framework with deterministic citations for agricultural knowledge access in developing regions. IR for Good track. DOI · arXiv

SycoLex: A Cross-Jurisdictional Benchmark for Sycophancy in Legal Case Reasoning CIKM 2026, under review

S. Banerji Seal et al. Resource track: 1,954 cases, three jurisdictions, six sycophancy-inducing prompt variants, five evaluated LLMs, judge and human-expert annotations.

Hierarchical Opinion Classification using Large Language Models FIRE 2025

S. Banerji Seal, A. Mishra, U. Ghosh. Parameter-efficient fine-tuning of Gemma with a two-layer classification head and class-weighted cross-entropy over an 8-class flattening of a three-level hierarchy. Paper

IISERK@ToT 2024: Query Reformulation and Layered Retrieval for Tip-of-the-Tongue Items TREC 2024

S. Banerji Seal, S. Adhikary, S. Sar, D. Roy. Four-step query reformulation with two-layer BM25 retrieval; Recall@1000 of 0.8067. Paper

Computational Modeling of [VO(SALIEP)(DTP)] as a Water-Reducing Catalyst To be submitted

S. Banerji Seal, guided by Dr. S. Roy. DFT (B3LYP) molecular-orbital energies in Gaussian with transition-state and reaction-pathway analysis.

Flagship Open-Source Software

Molecule3D / LAMMPS Web GUI 2026–

Browser-native molecular simulation workbench; MIT, GPU-accelerated, fully client-side

A complete **LAMMPS** workbench in the browser, built to lower the entry barrier to molecular dynamics. The **Script Builder** composes input decks as a drag-and-drop flowchart over a 186-command library covering the full general-command surface, locked by a coverage test against docs.lammps.org, and re-imports existing in.* scripts. The **Compiler Helper** turns 27 CMake options into ready-to-run build scripts. The **Structure Viewer** renders LAMMPS data, XYZ, PDB and CIF plus dump trajectories, with playback, MP4 capture, measurement tools and PBC-aware RDF, MSD and density analysis. **Stack:** TypeScript, React, Three.js / R3F, WebGL, Vitest (143 tests).

IISERKconnect – Campus Super-App 2026–

Sole developer. In campus-wide testing with 2,000+ users at IISER Kolkata

One Android (Kotlin) and web application unifying the entire IISER Kolkata intranet: WeLearn courses, files, deadlines and attendance; mess transactions, budget analytics and menu; academic calendar and events; grade cards; the question-paper archive; notice boards; library catalogue; the ePrints archive; VoIP directory; TCP counter and MAC registration; and a network dashboard. Each service is a hand-written HTML parser (Kotlin/Jsoup, ported 1:1 to DOMParser on web) behind captcha-backed logins and a campus-only allow-list, with credentials never leaving the device. Adds a serverless **Campus Chat** over encrypted WebRTC.

AgriIR Framework 2025–26

Reference implementation of the ECIR 2026 paper

Six-stage configurable RAG pipeline with deterministic citations and domain agents over FAISS, hybrid BM25 plus dense retrieval, and Ollama-served open-weight LLMs.

Introduction to Molecular Simulation 2026

Most-starred public repository. A beginner-to-running-simulation course: statistical mechanics, force fields, MD integrators and thermostats versus Monte Carlo, NVE/NVT/NPT ensembles, and observables (RDF, MSD, VACF, free energy).

DFT Notes 2026

Open knowledge base on density functional theory: the many-body Schrödinger problem through Hartree-Fock, the Hohenberg-Kohn theorems, the Kohn-Sham equations and exchange-correlation functionals, with runnable Python.

Fernholz Stochastic Portfolio Theory Library Ongoing

Diversity-weighted portfolios and relative arbitrage via stochastic calculus.

Research & Industrial Experience

Hybrid RAG Architecture for Verifiable Answer Synthesis

July 2025

Under Dr. Dwaipayan Roy, Computational and Data Sciences, IISER Kolkata

End-to-end RAG pipeline over IISER Kolkata intranet documents: Selenium acquisition, OCR with PyMuPDF and EasyOCR, context-aware chunking, Qwen3-4B embeddings in FAISS, hybrid dense plus BM25 retrieval fused by Reciprocal Rank Fusion, and a Streamlit interface returning verifiable inline citations.

Advanced Retrieval for TREC Tip-of-the-Tongue

2024

Guided by Dr. Dwaipayan Roy, IISER Kolkata

Multi-layer BM25 filtering in Lucene/Java with dynamic search-domain contraction, plus transformer-based query expansion via local LLMs using multi-shot and chain-of-thought prompting, reaching parity with modified DPR models.

Research Intern, HistoXai (Astroloop Technologies Pvt. Ltd.)

May-Jul 2025

Bangalore, India

Comparative analysis of 30+ open-source digital histopathology quality-assessment frameworks (HistoQC, PathProfiler, GrandQC, HistoROI, FASTPathology), evaluating architectures, datasets and metrics for downstream model integration. **Tech:** Python, OpenCV, PyTorch, Scikit-learn.

Entrepreneurial Ventures

Co-Founder & Lead Developer, iFiNN



2025–Present

AI-fintech startup, MeitY Startup Hub (GENESIS) funded, incubated at RISE Foundation, IISER Kolkata

Democratising ML and AI tooling through a simple interface for automated smart trading alerts across stocks, crypto and mutual funds. Funded; DPIIT registration in progress (Q1 2026); shipping by 2027.

Co-Founder & CTO, UnderWater AI



2025–Present

DeepTech startup, MeitY Startup Hub (GENESIS) funded, incubated at RISE Foundation, IISER Kolkata

Deep neural networks and early-fusion learning for underwater image enhancement and marine species identification. Funded; DPIIT registration in progress (Q1 2026); shipping by 2027.

Open Datasets Released

Hugging Face: huggingface.co/ShuvBan. Two public datasets; further datasets private or pre-release.

SycoLex



CC-BY-4.0

1,954 real legal cases across three jurisdictions (US Supreme Court 300, Indian Supreme Court 1,500, Indian Consumer Courts 154) with six prompt variants, five LLM response sets, and paired judge and expert annotations. The evaluation resource for the FIRE 2026 SYCO PHANCY shared task.

AgriIR_dataset



CC-BY-4.0

15,247 curated Indian agricultural knowledge entries with quality scores and domain metadata for crops, soils, regions and farming practices.

Interactive CV

3D CV Explorer



2026

This CV is also published as a walkable Three.js timeline: a 3D road along which every milestone appears in chronological order as an explorable station.

Invited Talks, Lectures & Jury Positions

Guest Lecturer & Guest of Honour, CBSE

District-Level Deliberation



Aug 2026

CBSE STEM DLD on Computational Thinking & AI, Abhinav Bharati High School, Kolkata District

Invited by the Central Board of Secondary Education as an **AI subject expert** to address **64 teachers from 38 schools** on “AI in the Real-World Context”: what today’s AI tools and services actually are, the next-token-prediction machinery underneath them, and how to teach with and about them responsibly. The full 60-minute talk is released openly as a 26-slide [HTML deck](#) and a matching [Beamer PDF](#).

Invited Judge, Voice of Youth 2026 National Summit

1–2 Aug 2026

Raajkutir, Kolkata; national teen-led innovation summit supported by Mr. Harshavardhan Neotia

Judged shortlisted student teams presenting in the technology and innovation domains, assessing technical depth, feasibility and demonstrated impact.

Track Co-organiser, FIRE 2026 SYCO PHANCY

Shared Task



2026

Explainable AI in Legal Reasoning: From Statute Prediction to Sycophancy Detection

Co-organising an international shared task with IISER Kolkata, Université de Bretagne Occidentale (France) and the University of Amsterdam.

Technical Talk, Running and Optimising Local LLMs

Aug 2025

Slashdot Student Chapter, IISER Kolkata

Hardware selection, quantisation, embedding pipelines, vector databases and practical optimisation for local LLM deployment.

Invited Talk, Sustainability, AI and Emerging Technologies

Aug 2025

Valence Chemistry Society, IISER Kolkata

Bridging sustainability science with AI-driven modelling, retrieval systems and computational chemistry tooling.

Domain Judge, Computer Science Events

July 2025

La Martiniere for Boys Annual Science Fest, Kolkata

External judge for project demonstrations, live coding and problem-solving rounds.

Leadership & Community Impact

President, Slashdot – Programming & Design Club



Aug 2025–

IISER Kolkata. *Elected with a 99% approval vote*

Elected to lead the official programming and design club with a 99% mandate, one of the strongest endorsements recorded for the post. Directing its workshops, hackathons, machine-learning and local-LLM sessions, and its open-source and web presence.

- **Office Bearer**, Valence Chemistry Club, IISER Kolkata (Aug 2025–).

- **Organiser**, Qiskit Fallfest 2025, sponsored by IBM Quantum (one of 150 institutes selected globally).

- **Event Lead**, Anicon 3.0 and 2.0 (500+ participants); Supra-Molecular Discussions 2024; GIAN courses on Soft-Oxometalates and X-Ray Crystallography.

- COVID-19 relief coordinator (2020–21); mentor with the Ek-Pehal education initiative.

- District-level debate and quiz finalist · Shotokan Karate · fourth-year art and painting student.

Technical Skills

Computational Chemistry & Materials: Density Functional Theory (DFT); **Quantum ESPRESSO** (plane-wave / pseudopotential SCF, relaxation, band structure, DOS, phonons); Gaussian (B3LYP, transition states); ORCA; VASP; Molecular Dynamics (LAMMPS, GROMACS, VMD, OVITO); force fields (AIREBO, ReaxFF, EAM, Lennard-Jones); Monte Carlo; exchange-correlation functionals; RDF, MSD and free-energy analysis

Machine Learning & Deep Learning: PyTorch, TensorFlow, transformers, mixture-of-experts, parameter-efficient fine-tuning (LoRA/QLoRA), distributed and mixed-precision training, Gaussian process regression, XGBoost, Scikit-learn; digital pathology (HistoQC, PathProfiler, GrandQC, HistoROI, FASTPathology)

LLMs, RAG & Agentic AI: retrieval-augmented generation, hybrid dense and sparse retrieval, FAISS, ChromaDB, LangChain, Ollama, vLLM, HuggingFace Transformers and Datasets, prompt engineering, autonomous agents, LLM-as-a-Judge, sycophancy and bias evaluation, AI safety; Qwen, Gemma, DeepSeek, LLaMA

Information Retrieval: Apache Lucene, BM25, Reciprocal Rank Fusion, query reformulation and expansion, evaluation (Recall@k, NDCG)

Bioinformatics & Formal Methods: RNA and protein structure prediction, sequence analysis, PyMol, ChimeraX, AutoDock Vina; Lean 4 and Mathlib, machine-checked theorem proving, formal verification

Languages: Python, C/C++, Kotlin, TypeScript/JavaScript, Java, Rust, Fortran, Lean 4, Bash, SQL

Web, Mobile & Systems: React, Three.js / React Three Fiber, WebGL and GLSL, Vite, Django, Flask, Node.js, WebRTC, PostgreSQL, MongoDB, SQLite3, Git, GitHub Actions (CI/CD), Docker, CMake, Linux, CUDA; Android (Kotlin), Jetpack Compose, Coroutines, Room, Jsoup; GTK4 in C, Qt in C++

Instrumentation: UV-Vis, ATR-FTIR, TGA, DSC, XRD, GC-MS, SEM, TEM, AFM, fluorimeter

Other: L^AT_EX, Beamer, Marp, Figma, technical writing

Education

IISER Kolkata 2022–2027 (exp.)
BS-MS Dual Degree, Chemistry Major, Computer Science Minor
CGPA 8.2

University of Calcutta 2021–2022
B.Sc. Honours in Physics (first year only) CGPA 8.308

Jodhpur Park Boys' High School (WBCHSE), Higher Secondary in Physics, Maths, Chemistry and Computer Science, 83% (2021) · **The New Horizon High School** (WBBSE), Secondary, 83.75% (2019)

Professional Experience

Freelance Web Developer & Designer 2024–2025
Built the site for ChemActiva Innovations Pvt. Ltd. (nanocellulose startup, DST-NIDHI PRAYAS grantee), Anicon 3.0 and the EFAML lab.

Private Educator & Technical Trainer 2018–Present
Self-employed, Kolkata. Computer science, physics and chemistry for ICSE, CBSE and WB Board students; 50+ mentored.

Technical Consultant 2021–Present
Self-employed, Kolkata. HPC solutions, system optimisation and BIOS/UEFI configuration; 50+ custom builds.

References available upon request

Achievements & Honours

Hackathons

- **UIDAI Data Hackathon 2026 – 1st Prize, Rs. 2,00,000** Jan 2026
UIDAI, NIC and MeitY. Won among 5,000+ submitting teams: Aadhaar enrolment and biometric-update patterns by region and demographic, with service-delivery recommendations.
- **Capital One Launchpad – Top 14 of 5,073 teams** 2025
IndicAgri, a multi-modal RAG advisor for Indian agriculture in 20+ languages, with a pipeline that curated a 15k-document dataset.
- **StatusCode1 – 1st Rank, GIAN Track** 2024
Natural-language search over GIAN's abandoned US patents with Nomic embeddings.
- **StatusCode0 – 1st Rank, MATLAB Track, IIIT Kalyani** 2023
- **Smart India Hackathon – Finalist** (SIH1701, SIH1734) 2024

Competitions, Examinations & Honours

- **ChemEnigma – 1st Rank**, 72-hour chemistry contest, IISc Bangalore (2025)
- **All Bengal Chemistry Quiz – 2nd Runners Up**, Presidency University (2025)
- **Mimansa – Zonal Topper**, mathematics, IISER Pune (2024)
- **NAEST – Zonal Runners Up**, IIT Kanpur and Shiksha Sopan (2023)
- **JEE Mains & Advanced** top 0.1% · **IAT** top 0.06% · **WBJEE** top 0.05% (2022)
- **Reliance Foundation Undergraduate Scholar** (2023), top 5,000 nationally
- **Best Young Scientist Speaker on Nanotechnology** (2019), World Science Conference, Jadavpur Univ.

Selected Projects

Solvation of C₆₀ Nanoparticles in Water 2025
LAMMPS molecular dynamics with AIREBO potentials, automated feature extraction (RDF, coordination numbers), and a surrogate ML pipeline (GPR, neural nets, XGBoost).

opngx – High-Speed Camera Footage Extractor 2026
Pixel-exact extractor for Optronis .bin recordings saturating all CPU cores with runtime SIMD dispatch; CLI, GUI and Python library.

Legal Document Retrieval RAG · Agentic Database Builder 2024–25
Retrieval over the Commercial Courts Act corpus with LLaMA-3.2, ChromaDB and Mistral-OCR; and LLM-led schema generation and curation.

WeLearn Bot in C · E-Commerce GUI in C 2024–25
Multi-threaded course-resource automation with CLI and GTK4 GUI; and a GTK4 + SQLite3 storefront with a BM25-backed chatbot.

Twists Simulator · Spherical Harmonics · TCP Monitor · Wi-Fi Optimizer 2023–25
Ray-Dutt and Bailar twists in interactive 3D; a Flask/Plotly quantum orbital visualiser; systemd-integrated TCP logging; automated Wi-Fi channel selection; and a Django rewrite of the IISER-K canteen system serving 2,000+ users.

Published Author, MindScapes (ISBN 978-9389923209) 2020
Creative anthology; workshops on technical and creative writing.

Open Curricula 2025–26
A 20-module C course, a Python course through to local LLM deployment, eleven GPU-shader lessons, and CH4114 tutorials.